

Structural and Thermal Design Optimisation of a High-Performance Smartphone Chassis Through Finite Element Analysis

Problem Statement & Objectives

In November 2024, **91% of American adults owned a smartphone** – up from just 35% in 2011. However, this rapid growth has not come without design challenges. Even some of the largest manufacturers have produced products that, over time, were deemed defective:

- **Apple’s “Bendgate** saw users report the **iPhone 6 and 6 Plus for bending** in their pockets under relatively low loading conditions.
- **Samsung’s Galaxy Note 7** was recalled after their attempts to decrease battery size and increase power capacity led to **device explosions**.

Therefore, the goal of this project was to design a phone chassis that:

1. Is **lightweight**
2. Is **aesthetically attractive**
3. Is **stiff and strong** enough to withstand bending and torsion loads
4. Uses a realistic and **suitable material**
5. Efficiently **dissipates heat** generated by the phone's CPU and battery

Material Selection

After conducting research, four candidate materials were evaluated based on structural performance, thermal conductivity, and manufacturability.

Property	Ti-6Al-4V	Al 6061-T6	AZ91D	CFRTP
Density (kg/m³)	4430	2700	1810	1650
Young's Modulus (GPa)	113.8	68.9	44.8	24
Thermal Conductivity (W/m·K)	6.7	167	72.7	8
Specific Stiffness (GPa·cm³/g)	25.7	25.5	24.8	15

Eliminated ✗ or Shortlisted — :

- | | |
|--|--|
| <p>✗ Titanium:</p> <p>✗ Carbon Fibre:</p> <p>— Magnesium (AZ91D):</p> | <ul style="list-style-type: none"> ▪ High density, poor thermal conductivity. ▪ Anisotropic behaviour, poor thermal conductivity. ▪ Lowest density, moderate thermal performance. |
|--|--|

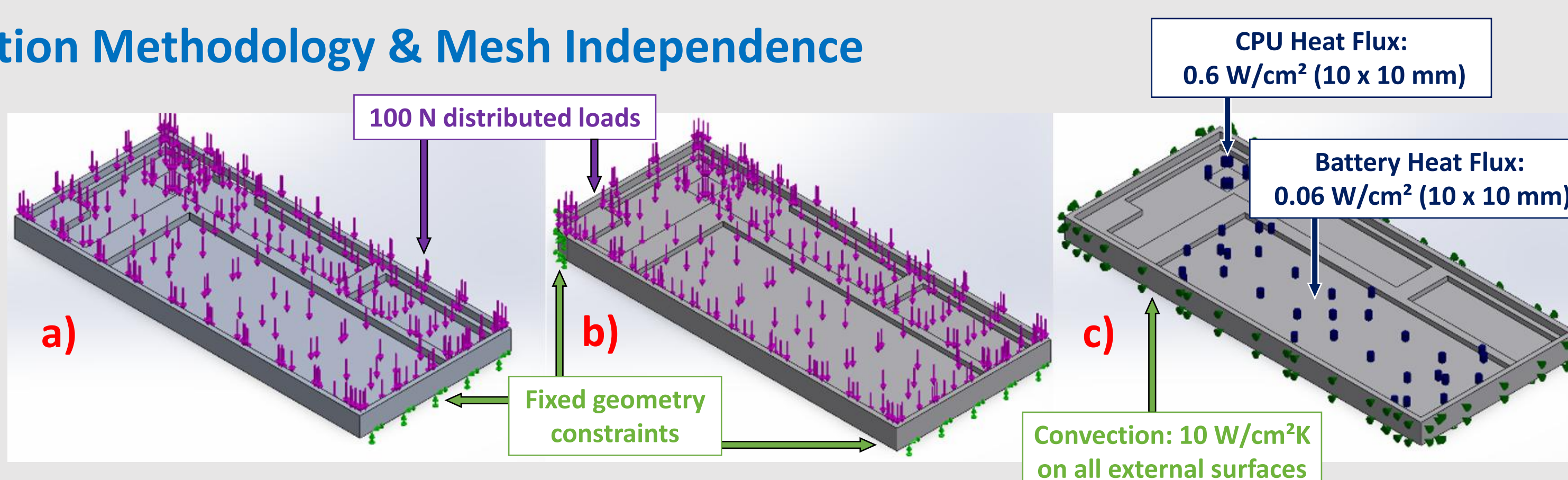
Selected ✓:

- | | |
|--------------------------|---|
| ✓ Aluminium 6061: | Best thermal conductivity, higher stiffness, lower strength ratio, can be anodised to improve thermal performance, better corrosion resistance. |
|--------------------------|---|

Simulation Methodology & Mesh Independence

To begin, a 160 x 70 x 9 mm simplified chassis was used to isolate mesh effects.

- a) Simple bending setup
- b) Torsion setup
- c) Thermal heat flux setup

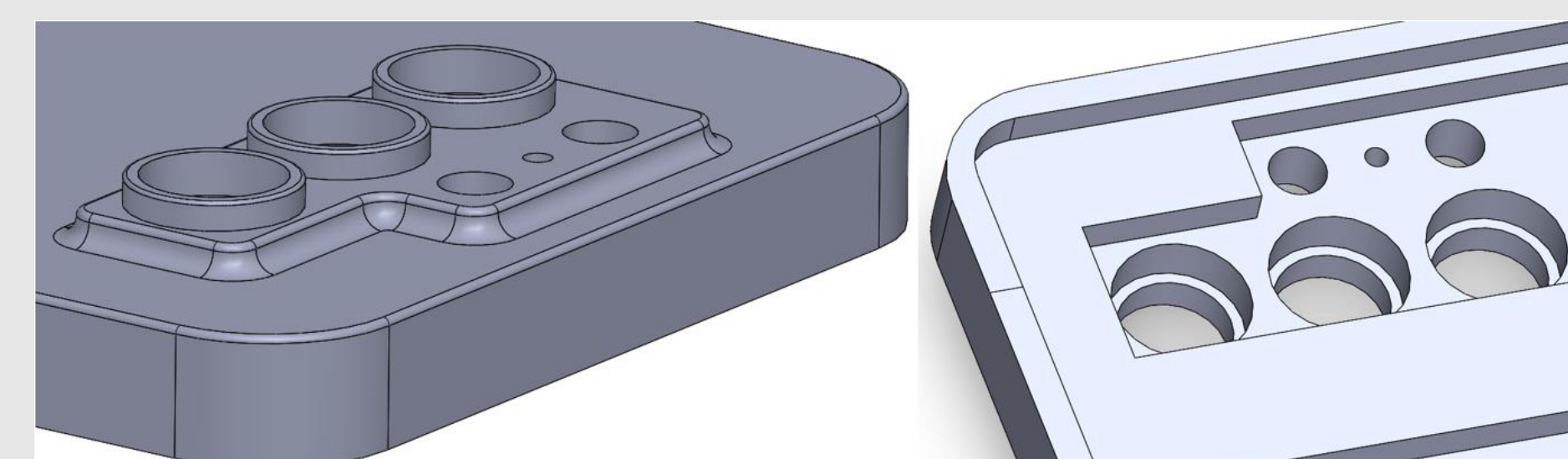


Structural Mesh Independence

Case A: Simple Bending	Case B: Torsion & Bending
Problem: - Global mesh refinement caused stress singularities at fixed edges - Max stress diverged: 14.36 MPa → 28.86 MPa (48.6% error) - Displacement and strain energy converged, stress did not	Method: - H-adaptive refinement used to automatically refine elements in high-error regions. - Target accuracy: 98% 5 refinement iterations carried out - Curvature-based meshing to capture geometry detail.
✓ Solution: - Local mesh control was applied along the chassis frame. 4 mm global + 0.8 mm local.	✓ Converged: - Displacement: 0.0645 mm (0.28%) - Strain energy: 1.499E-03 J - Strain energy norm error: 2.13% (just over 2% target)
✓ Result: - Stress converged at 14.36 MPa - Displacement: 0.085 mm 0% relative error.	✗ Did Not Converge: - Stress singularity at fixed corner (unavoidable) - Max stress differential: 125 → 2900 MPa

Parametric Design Optimisation

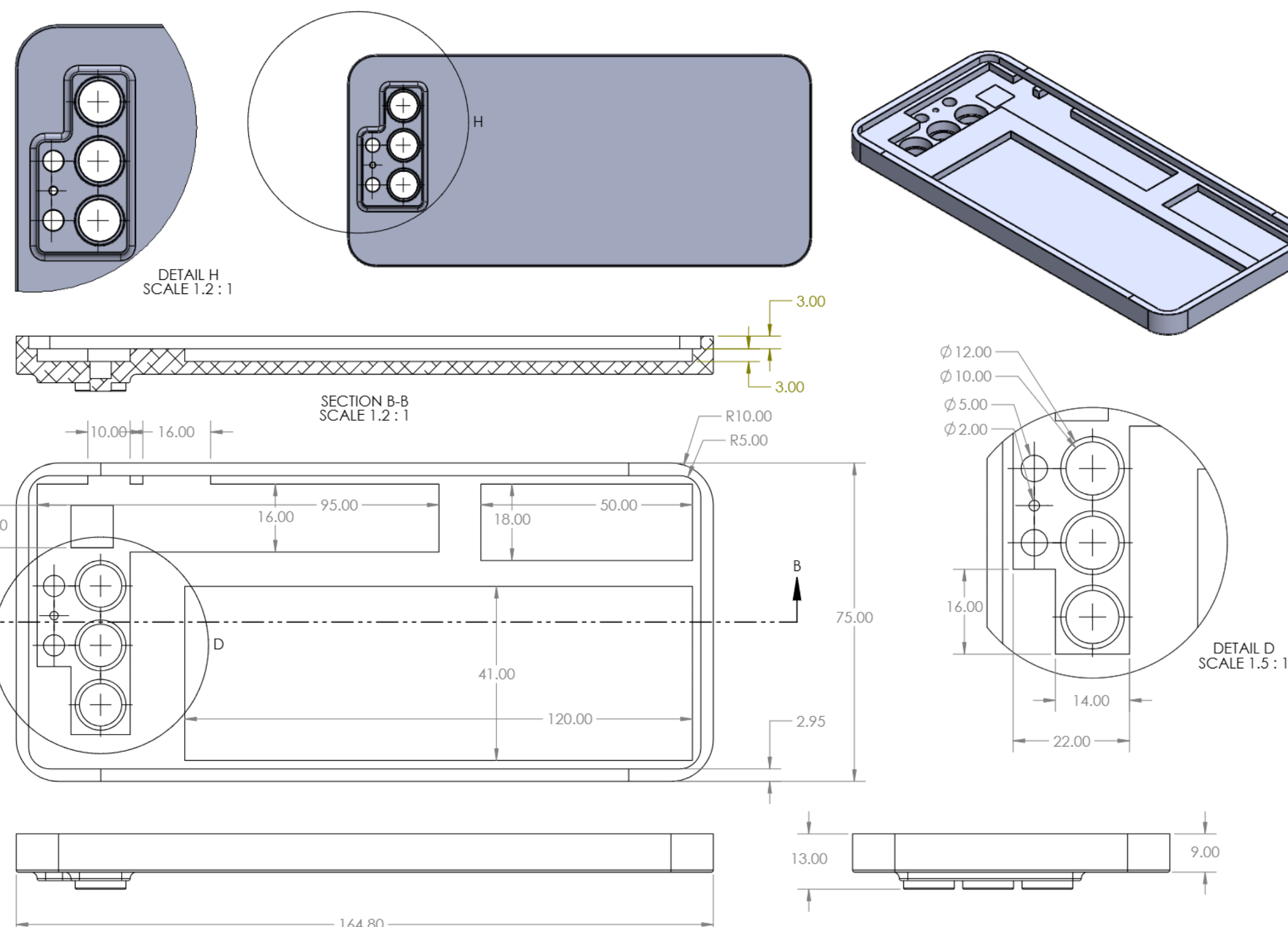
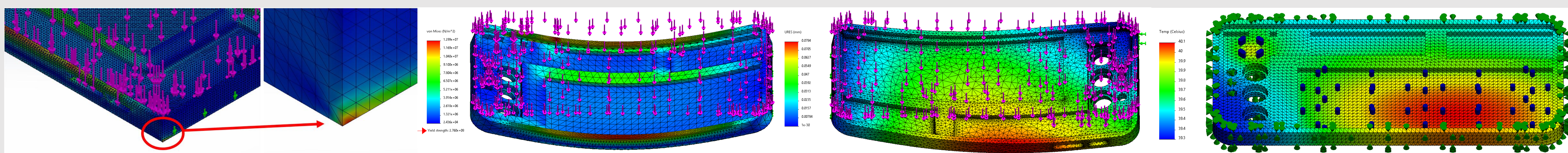
Variables: ▪ Height: 156 – 165 mm ▪ Width: 70 – 75 mm ▪ Depth: 8 – 9 mm ▪ Frame Thickness: 1 – 3 mm	Constraints: ▪ Stress: $\leq 0.6 \times \sigma_{yield}$ ▪ Displacement: ≤ 1.6 mm
Goals: ▪ Minimise temperature ▪ Minimise mass	
Optimal Dimensions: 164.8 x 75 x 9 mm Frame: 2.95 mm	



Thermal Mesh Independence

- Refined: 4 mm $\rightarrow$ 2 mm $\rightarrow$ 1 mm
- Max temp: 43.5°C (constant across all meshes)
- 0% relative error

✓ **Adopted:** 2 mm global (accuracy with low compute cost)



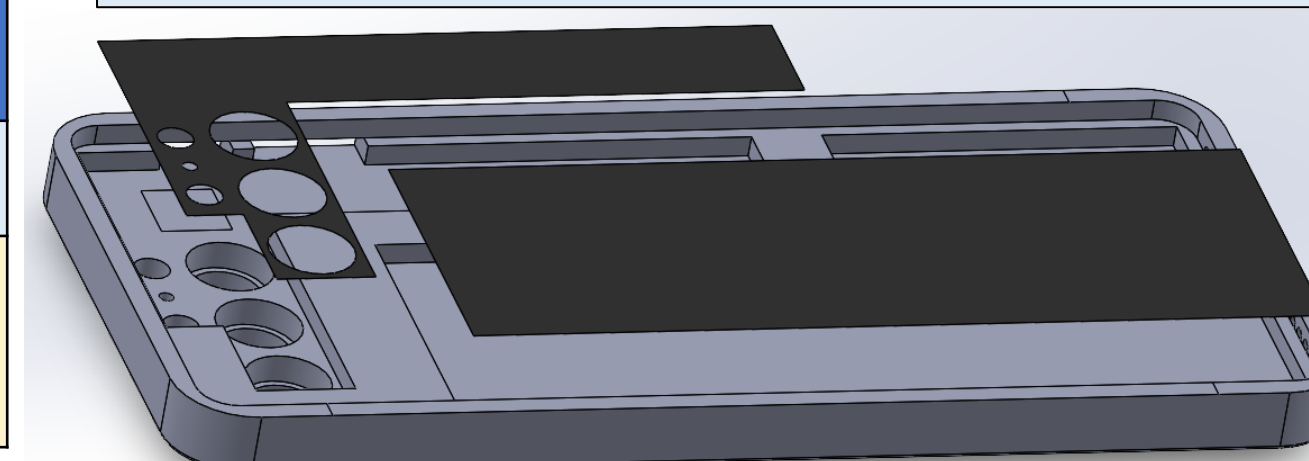
Final Design Simulation Performance

Simple Bending	Torsion	Thermal
▪ Max δ: 0.0802 mm ▪ Max σ: 12.99 MPa	▪ Max τ: 0.0784 mm ▪ Stiffness: 1,275 N/mm	▪ Min temp: 34.8°C ▪ $R_{thermal}$: 4.3 K/W

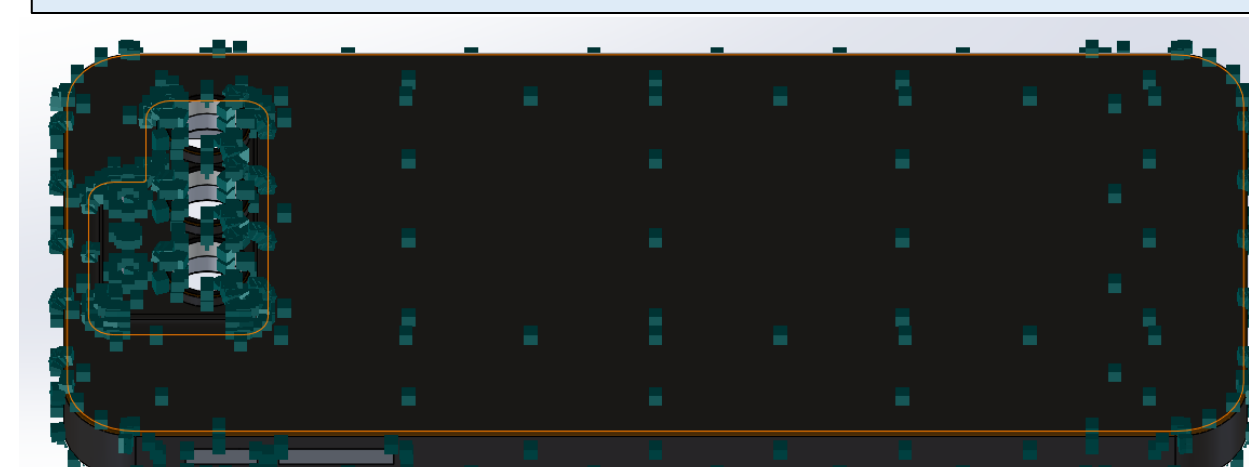
Results and Thermal Enhancements

Structural Performance	Thermal Optimisation
▪ Stiffness: 1,247 N/mm ▪ Specific Stiffness: 8.7 (N/mm)/g ▪ Mass: 143.5 g ▪ Strength ratio: 4.7% ▪ Buckling FOS: 400 approx.	▪ Baseline: 40.1°C + Graphene Layer → 39.8°C + Anodised Finish → 35.5°C + Vent perforations → 35.2°C ▪ Max temp: 35.2°C (safe) ▪ Total ΔT: -4.9°C

Graphene Layer (heat dispersion)



Anodised Finish (heat dissipation)



Vent Perforations (heat convection)

